

Lecture 3: Introduction to Information Theory and Uncertainty Quantification

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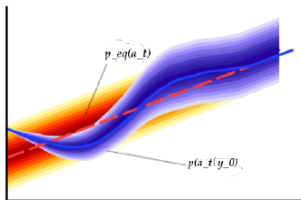
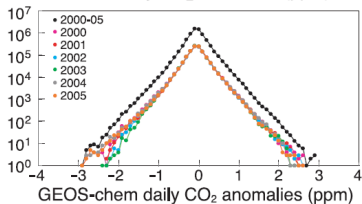
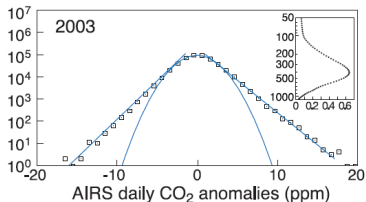
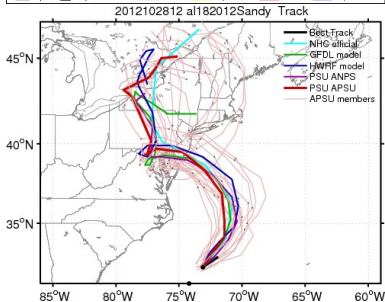
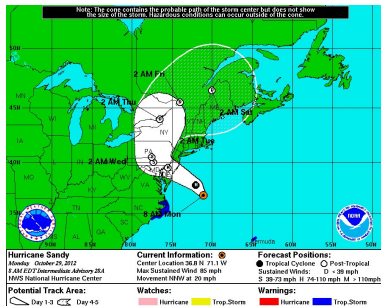
Introduction to Data Assimilation
2026 Spring



Outline

1. Entropy
2. Relative entropy
3. Mutual information

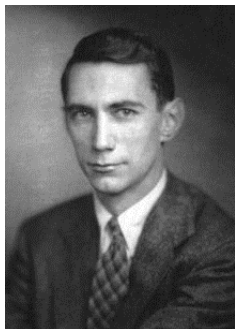
Some basic examples ...



Some important questions in uncertainty quantification.

- ▶ How can we measure our level of uncertainty about a physical system?
- ▶ What is the lack of information of an imperfect model relative to nature?
- ▶ How much information can we expect to gain about a physical system when we perform a measurement/observation?

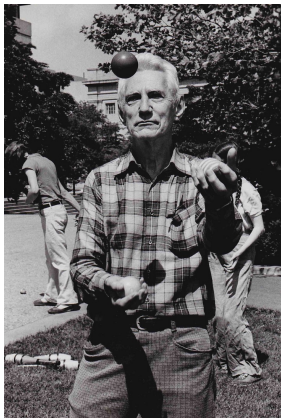
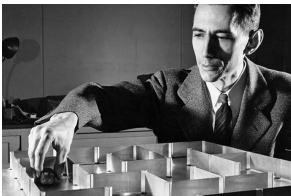
Claude Elwood Shannon



- ▶ Claude Elwood Shannon (1916 – 2001) was an American mathematician, electronic engineer, and cryptographer known as “the father of information theory”.
- ▶ Pioneered by Shannon in a 1948 article, “[A mathematical theory of communication](#)”, *Bell System Technical Journal*.
- ▶ The article was later published in 1949 as a book by Shannon and Warren Weaver, “[The Mathematical Theory of Communication](#)”, Univ of Illinois Press.
- ▶ Shannon's article in 1951 [Prediction and Entropy of Printed English](#) was the contribution of information theory to natural linguistics.

Shannon's other work, hobbies and inventions,

- ▶ Shannon's mouse (machine learning!)
- ▶ Computer chess program
- ▶ Juggling
- ▶ Using information theory to calculate odds while gambling ...
- ▶ Ultimate machine (the greatest invention!)
- ▶ ...



Outline

- ▶ Basic concepts of information theory and the Shannon entropy

1) The Shannon entropy, 2) uniqueness, 3) maximum entropy principle, 4) coarse graining and loss of information, 5) Gaussian framework and the relation with the RMS error

- ▶ The relative entropy

1) The relative entropy, 2) quantifying the lack of information, 3) empirical theory for model selection, 4) estimating errors in posterior distribution, 5) Gaussian framework

- ▶ The mutual information

1) The mutual information, 2) superensemble expectation, 3) Gaussian framework and the relation with the pattern correlation

Basic concepts of information theory and the Shannon entropy

1. Shannon's intuition from the theory of communication.

Representing a “word” in a message as a sequence of binary digits with length n .

- ▶ The set $\mathcal{A}_{2^n}$ of all words of length n has $2^n = N$ elements.
- ▶ The amount of information needed to characterize one element is $n = \log_2 N$.

Example:

$n = 7, N = 128.$

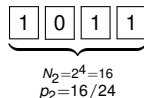
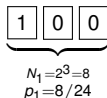
1	0	1	0	1	0	1
0	1	1	1	0	1	0
⋮	⋮	⋮	⋮	⋮		

Following this type of reasoning, the amount of information needed to characterize an element of any set, $\mathcal{A}_N$, is $n = \log_2 N$ for general N .

- ▶ Now consider a situation of a set $\mathcal{A} = \mathcal{A}_{N_1} \cup \dots \cup \mathcal{A}_{N_k}$, where the set $\mathcal{A}_{N_i}$ are pairwise disjoint from each other with $\mathcal{A}_{N_i}$ having N_i total elements.
- ▶ Set p_i to be given by $p_i = N_i/N$, where $N = \sum N_i$.

Example: $\mathcal{A} = \mathcal{A}_{N_1} \cup \mathcal{A}_{N_2}$

$N = 24$, $N_1 = 8$, $N_2 = 16$.



- ▶ If we know that an element of $\mathcal{A}$ belongs to some $\mathcal{A}_{N_i}$, then we need $\log_2 N_i$ additional information to determine it completely.
- ▶ The **average amount of information** we need to determine an element, provided that we already know the $\mathcal{A}_{N_i}$ to which it belongs, is given by

$$\sum_i \frac{N_i}{N} \log_2 N_i = \sum_i \frac{N_i}{N} \log_2 \left(\frac{N_i}{N} \cdot N \right) = \sum p_i \log_2 p_i + \log_2 N.$$

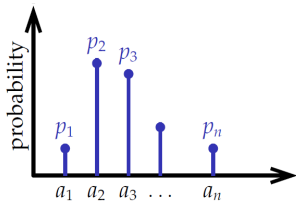
- ▶ Recall that $\log_2 N$ is the information that we need to determine an element given the set $\mathcal{A}$ if we do not know to which $\mathcal{A}_{N_i}$ a given element belongs.
- ▶ Thus, the corresponding average lack of information is

$$- \sum p_i \log_2 p_i.$$

2. Definition of the Shannon entropy (discrete case)

Let p be a finite, discrete probability measure on the sample space $\mathcal{A} = \{a_1, \dots, a_n\}$

$$\mathcal{PM}_n(\mathcal{A}) = \left\{ p = \sum_{i=1}^n p_i \delta_{a_i}, \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1 \right\}.$$



The Shannon entropy $\mathcal{S}(p)$ of the probability p is defined as

$$\mathcal{S}(p) = \mathcal{S}(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \ln p_i.$$

3. Uniqueness of the Shannon entropy

Let H_n be a function defined on the space of discrete probability measure $\mathcal{PM}_n(\mathcal{A})$ and satisfying the following properties

1. $H_n(p_1, \dots, p_n)$ is a continuous function.
2. $A(n) = H_n(1/n, \dots, 1/n)$ is monotonic increasing in n , i.e., H_n increases with increasing uncertainty,

$$H_n(\{1/n, \dots, 1/n\}) > H_{n'}(\{1/n', \dots, 1/n'\}), \quad n > n'$$

3. Composite rule:

$$\begin{aligned} H_n(p_1, \dots, p_k, p_{k+1}, \dots, p_n) &= H_2(w_1, w_2) + w_1 H_k(p_1/w_1, \dots, p_k/w_1) \\ &\quad + w_2 H_{n-k}(p_{k+1}/w_2, \dots, p_n/w_2), \end{aligned}$$

where $w_1 = p_1 + \dots + p_k$, $w_2 = p_{k+1} + \dots + p_n$ and p_i/w_j is the conditional probability.

Then H_n is a positive multiple of the Shannon entropy

$$H_n(p_1, \dots, p_n) = KS(p_1, \dots, p_n) = -K \sum_{i=1}^n p_i \ln p_i, \quad \text{with } K > 0.$$

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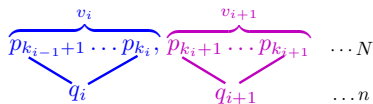
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Key of proof: Using composite rule $\implies A(nv) = A(n) + A(v)$.
The only function satisfying this condition is $A(n) = K \ln n$.

Sketch of proof.

- Due to (1), only need to prove for rational $q_1, \dots, q_n$ with $\sum_{i=1}^n q_i = 1$.
Set $q_i = v_i/N$, $N = \sum_{i=1}^n v_i$.
- Consider sample space $\{p_1 \dots p_N\}$ and partition it into n disjoint subsets.

- $H_n(p_1, \dots, p_n)$ is continuous.
- $A(n) = H_n(1/n, \dots, 1/n)$ is monotonic increasing in n .
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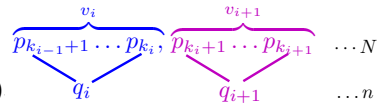
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- Consider sample space $\{p_1 \dots p_N\}$ and partition it into n disjoint subsets.
- By composite rule (3),

$$H_N(p_1, \dots, p_N) = H_n(q_1, \dots, q_n) + \sum_{i=1}^n q_i H_{v_i}(p_{k_{i-1}+1}/q_i, \dots, p_{k_i}/q_i)$$

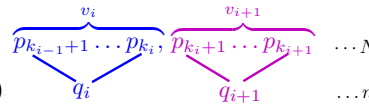


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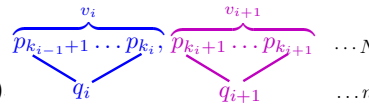
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$$A(nv) = A(n) + A(v).$$

The only continuous function $A(n)$ satisfying this condition is $A(n) = K \ln n$.

- Inserting this back to $A(N)$ equation leads to

$$H_n(q_1, \dots, q_n) = K \ln n - K \sum_{i=1}^n q_i \ln n_i = K \sum_{i=1}^n q_i \ln \frac{1}{q_i}$$

4. Maximum entropy principle

- ▶ With a given probability measure $p \in \mathcal{PM}(\mathcal{A})$, various statistical measurements can be made with respect to this probability measure p . The expected value, or statistical measurement, of f with respect to p is given by

$$\langle f \rangle_p = \sum_{i=1}^n f(a_i) p_i.$$

- ▶ A practical issue is to look for the least biased probability distribution consistent with certain statistical constraints,

$$\mathcal{C}_L = \{p \in \mathcal{PM}(\mathcal{A}) | \langle f_j \rangle_p = F_j, 1 \leq j \leq L\}.$$

The least biased probability distribution p_L given the constraints $\mathcal{C}_L$ is the maximum entropy distribution,

$$\max_{p \in \mathcal{C}_L} \mathcal{S}(p) = \mathcal{S}(p_L), \quad p_L \in \mathcal{C}_L.$$

- ▶ Uncertainty decreases with more information being included¹.

$$\mathcal{S}(p_L) \leq \mathcal{S}(p_{L'}), \quad \text{for } L' < L.$$

¹In thermodynamics, entropy is commonly associated with the amount of order, disorder, or chaos in a thermodynamic system.

Example. Find the least biased probability distribution p on $\mathcal{A} = \{a_1, \dots, a_n\}$ with no additional constraints.

$$\text{Maximize} \quad \mathcal{S}(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \ln p_i,$$

$$\text{Subject to} \quad \sum_{i=1}^n p_i = 1 \quad \text{and all } p_i \geq 0.$$

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$$\begin{array}{ll} \text{Maximize} & \mathcal{S}(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \ln p_i, \\ \\ \text{Subject to} & \sum_{i=1}^n p_i = 1 \quad \text{and all } p_i \geq 0. \end{array}$$

Construct the Lagrange function

$$L = - \sum_{i=1}^n p_i \ln p_i + \lambda \sum_{i=1}^n p_i.$$

The minimum p^* satisfies

$$\frac{\partial L}{\partial p_i} = 0, \quad i = 1, \dots, n,$$

which results in

$$\ln p_i^* + 1 + \lambda = 0, \quad i = 1, \dots, n.$$

This implies all the probabilities p_i^* are equal and therefore

$$p_i^* = 1/n.$$

Uniform distribution!

Example. Find the least biased probability distribution p on $\mathcal{A} = \{a_1, \dots, a_n\}$ subject to the $r + 1$ constraints ($r + 1 \leq n$),

$$F_j = \langle f_j \rangle_p = \sum_{i=1}^n f_j(a_i) p_i, \quad j = 1, \dots, r, \quad \sum_{i=1}^n p_i = 1.$$

Construct the Lagrange function

$$L = - \sum_{i=1}^n p_i \ln p_i + \lambda_0 \sum_{i=1}^n p_i + \sum_{j=1}^r \lambda_j \langle f_j \rangle_p.$$

Componentwise it yields a system of n equations for the unknowns p_i^*

$$\ln p_i^* = - \sum_{j=1}^r \lambda_j f_j(a_i) - (\lambda_0 + 1), \quad i = 1, \dots, n,$$

and solving for p_i^* we obtain

$$p_i^* = \exp \left(- \sum_{j=1}^r \lambda_j f_j(a_i) - (\lambda_0 + 1) \right).$$

Exponential family!

To eliminate the multiplier λ_0 , we utilize the constraint that the sum of all the probabilities p_i^* is 1. This simplifies the formula for p_i^* ,

$$p_i^* = \frac{\exp \left(- \sum_{j=1}^r \lambda_j f_j(a_i) \right)}{\sum_{i=1}^n \exp \left(- \sum_{j=1}^r \lambda_j f_j(a_i) \right)}.$$

The other Lagrange multipliers λ_i must be determined through the constraint equations (generally a non-trivial task).

5. Continuous case.

So far, we have discussed the Shannon entropy in the discrete case,

$$\mathcal{S}(\rho) = - \sum_{i=1}^n p_i \ln p_i.$$

It is easy to generalize it to the continuous measures.

- ▶ A continuous probability density, $\rho(\lambda)$ satisfies the two requirements

$$\rho(a) \geq 0, \quad \int \rho(a) da = 1.$$

- ▶ The continuous form of the Shannon entropy is given by

$$\mathcal{S}(\rho) = - \int \rho \ln(\rho) da.$$

- ▶ Maximum entropy principle holds for the continuous case as well.

Example: Find the least biased distribution given the constraints $\mathcal{C}$ defined by the first and second moments,

$$\begin{aligned} \rho(\lambda) &\geq 0, & \int_{\mathcal{R}^1} \rho(\lambda) d\lambda &= 1, \\ \bar{\lambda} &= \int_{\mathcal{R}} \lambda \rho(\lambda) d\lambda, & \sigma^2 &= \int_{\mathcal{R}^1} (\lambda - \bar{\lambda})^2 \rho(\lambda) d\lambda. \end{aligned}$$

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The variational derivative of the entropy is given by

$$\frac{\delta S}{\delta \rho} = -(1 + \ln \rho).$$

The first and second moment constraints are linear functionals of the density ρ so that

$$\frac{\delta \bar{\lambda}}{\delta \rho} = \lambda, \quad \frac{\delta \sigma^2}{\delta \rho} = (\lambda - \bar{\lambda})^2.$$

From the Lagrange multiplier principle we have, at the entropy maximum

$$-\frac{\delta S}{\delta \rho}|_{\rho=\rho^*} = -\mu_0 - \mu_1 \frac{\delta \bar{\lambda}}{\delta \rho}|_{\rho=\rho^*} - \mu_2 \frac{\delta \sigma^2}{\delta \rho}|_{\rho=\rho^*},$$

where μ_0, μ_1, μ_2 are the Lagrangian multiplier for the constraints.

Therefore

$$\ln \rho^* = (-\mu_0 + 1) - \mu_1 \lambda - \mu_2 (\lambda - \bar{\lambda})^2,$$

which defines ρ^* as a Gaussian probability density with mean $\bar{\lambda}$ and variance σ^2 so that $\rho^* = \rho_{\bar{\lambda}, \sigma}(\lambda)$.

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$$\rho(\lambda) \geq 0, \quad \int_{\mathcal{R}^1} \rho(\lambda) d\lambda = 1,$$

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From the Lagrange multiplier principle we have, at the entropy maximum

$$-\frac{\delta S}{\delta \rho}|_{\rho=\rho^*} = -\mu_0 - \mu_1 \frac{\delta \bar{\lambda}}{\delta \rho}|_{\rho=\rho^*} - \mu_2 \frac{\delta \sigma^2}{\delta \rho}|_{\rho=\rho^*},$$

where μ_0, μ_1, μ_2 are the Lagrangian multiplier for the constraints.

Therefore

$$\ln \rho^* = (-\mu_0 + 1) - \mu_1 \lambda - \mu_2 (\lambda - \bar{\lambda})^2,$$

which defines ρ^* as a Gaussian probability density with mean $\bar{\lambda}$ and variance σ^2 so that $\rho^* = \rho_{\bar{\lambda}, \sigma}(\lambda)$.

This fact supports the idea well known from **the central limit theorem** that **Gaussian densities are the most universal distributions with given first and second moments.**

Example. The least biased probability distribution p subject to the $r + 1$ constraints,

$$F_j = \langle f_j \rangle_p = \int f_j(a) p(a) da, \quad j = 1, \dots, r, \quad \int p(a) da = 1,$$

is given by

$$p^*(a) = \frac{\exp\left(-\sum_{j=1}^r \lambda_j f_j(a)\right)}{\int \exp\left(-\sum_{j=1}^r \lambda_j f_j(a)\right) da}.$$

Note that such exponential family of functions will be revisited later.

Example. Finite-moment problem for probability measures.

We quantify the information contained in the moments of a probability density $p(\lambda)$

$$\bar{\lambda} = M_1 = \int \lambda p(\lambda) d\lambda; \quad M_j = \int (\lambda - \bar{\lambda})^j p(\lambda) d\lambda, \quad 2 \leq j \leq 2L,$$

in addition to $\int p(\lambda) d\lambda = 1$. The probability distribution $p_{2L}^*(\lambda)$ with the least bias given these moment constraints, i.e., the one that maximizes the entropy, is:

$$\mathcal{S}(p_{2L}^*) = \max_{p \in PM_{2L}} \mathcal{S}(p).$$

This gives

$$p_{2L}^* = \exp \left(\sum_{m=0}^{2L} \alpha_m (\lambda - \bar{\lambda})^m \right),$$

where λ_m are the appropriate Lagrange multipliers. Note that we need even powers and $\alpha_{2L} < 0$ to guarantee a finite probability measure.

Since adding more moments increases information, it follows that

$$\mathcal{S}(p_{2L_2}^*) \leq \mathcal{S}(p_{2L_1}^*), \quad \text{for } L_1 \leq L_2.$$

$$\begin{aligned} -\frac{\delta \mathcal{S}}{\delta \rho} \Big|_{\rho=p^*} &= -\mu_0 - \mu_1 \frac{\delta M_1}{\delta \rho} \Big|_{\rho=p^*} - \mu_2 \frac{\delta M_2}{\delta \rho} \Big|_{\rho=p^*} - \dots - \mu_{2L} \frac{\delta M_{2L}}{\delta \rho} \Big|_{\rho=p^*}, \\ \ln p^* &= (-\mu_0 + 1) - \mu_1 \lambda - \mu_2 (\lambda - \bar{\lambda})^2 - \dots - \mu_{2L} (\lambda - \bar{\lambda})^{2L}. \end{aligned}$$

When is a probability distribution significantly non-Gaussian?

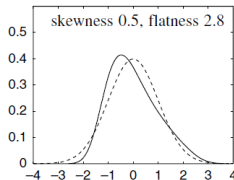
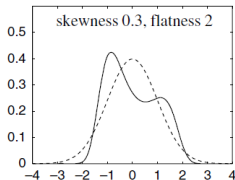
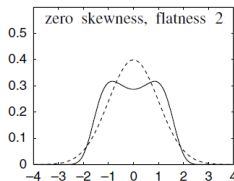
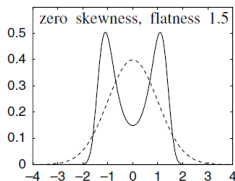
How much additional information is contained in the third and forth moments of a probability distribution beyond the Gaussian estimate, p_G^* ?

Here, we compare $\mathcal{P}(p_4^*, p_G^*)$ for varying values of the third and forth moments with fixed mean and variance.

The third and forth moments of p_4^* are characterized by the skewness and flatness (kurtosis)

$$\text{Skew} = \frac{\int (\lambda - \bar{\lambda})^3 p_4^*}{(\int (\lambda - \bar{\lambda})^2 p_4^*)^{3/2}},$$

$$\text{Flat} = \frac{\int (\lambda - \bar{\lambda})^4 p_4^*}{(\int (\lambda - \bar{\lambda})^2 p_4^*)^2}.$$



$\mathcal{P}(p_4, p_G)$	$Skew_{p_4} = 0$	$Skew_{p_4} = 0.3$	$Skew_{p_4} = 0.5$
$Flat_{p_4} = 1.5$	0.2772	0.3921	0.6921
$Flat_{p_4} = 2$	$6.015 \cdot 10^{-2}$	$9.879 \cdot 10^{-2}$	0.1885
$Flat_{p_4} = 2.8$	$1.016 \cdot 10^{-3}$	$1.226 \cdot 10^{-2}$	$3.842 \cdot 10^{-2}$

Multivariate Gaussian distribution

$$p_{\mathbf{x}}(x_1, \dots, x_k) = \frac{1}{\sqrt{(2\pi)^k |\mathbf{\Sigma}|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right),$$

where $\boldsymbol{\mu}$ is the k -dimensional mean and $k \times k$ matrix $\mathbf{\Sigma}$ is the covariance, which is always positive definite.

The **marginal distribution** of Gaussian over a subset of components is obtained by simply **dropping the remaining components**, e.g. given a Gaussian vector $(x_1, x_2, x_3)^T$ with $\boldsymbol{\mu} = (\mu_1, \mu_2, \mu_3)^T$ and covariance

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{pmatrix},$$

the marginal density for (x_1, x_3) is given by a Gaussian with

$$\tilde{\boldsymbol{\mu}} = \begin{pmatrix} \mu_1 \\ \mu_3 \end{pmatrix}, \quad \tilde{\mathbf{\Sigma}} = \begin{pmatrix} \Sigma_{11} & \Sigma_{13} \\ \Sigma_{31} & \Sigma_{33} \end{pmatrix}.$$

6. Coarse graining and loss of information

- ▶ Coarse graining is the terminology utilized in statistical physics for processes that involve averaging over small scales.
- ▶ Intuitively, coarse graining means that **information is lost so that the entropy should increase.**

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Rigorous statement for probability measures on the line.

Let $\lambda_i < \lambda_{i+1}$ for integers i , $-\infty < i < \infty$ with equi-distance spacing $\Delta\lambda$ and a probability measure $\rho(\lambda)$ define the discrete coarse-grained probability measure p by

$$p = \sum_i \rho_i \delta_{\lambda_i}, \quad \text{with} \quad \rho_i = \int_{\lambda_i}^{\lambda_{i+1}} \rho(\lambda) d\lambda.$$

- ▶ We expect that p contains **less** information than $\rho(\lambda)$.
- ▶ This intuition is confirmed by the following general inequality for $\Delta\lambda \leq 1$

$$\mathcal{S}(p) \geq \mathcal{S}(\rho) + \ln((\Delta\lambda)^{-1})$$

where

$$\mathcal{S}(p) = - \sum \rho_i \ln \rho_i, \quad \text{and} \quad \mathcal{S}(\rho) = - \int \rho \ln \rho.$$

Proof.

$$\begin{aligned} \mathcal{S}(\rho) &\geq \mathcal{S}(\rho) + \ln((\Delta\lambda)^{-1}), \\ \rho &= \sum_i \rho_i \delta_{\lambda_i}, \quad \rho_i = \int_{\lambda_i}^{\lambda_{i+1}} \rho(\lambda) d\lambda, \\ \mathcal{S}(\rho) &= - \sum \rho_i \ln \rho_i, \quad \mathcal{S}(\rho) = - \int \rho \ln \rho. \end{aligned}$$

We use the elementary inequality

$$x(\ln(x) - \ln(y)) \geq x - y.$$

Rewrite it in the more useful form

$$x \ln(x) \geq x \ln(y) + x - y.$$

Set $y = \rho_i / \Delta\lambda$ and $x = \rho(\lambda)$ and integrate the above equation over the interval $(\lambda_i, \lambda_{i+1})$,

$$\int_{\lambda_i}^{\lambda_{i+1}} \rho(\lambda) \ln \rho(\lambda) d\lambda \geq \rho_i \ln \rho_i - \rho_i \ln(\Delta\lambda).$$

Summing the above inequalities over i finishes the proof (note the negative sign in the Shannon entropy).

The elementary inequality is valid according to mean value theorem

$$\frac{\ln(x) - \ln(y)}{x - y} = \frac{1}{\xi} \geq \frac{1}{x}, \quad \text{for } x > \xi > y,$$

7. Gaussian framework

- Recall the Shannon entropy

$$\mathcal{S}(\rho) = - \int \rho \ln(\rho).$$

If $\rho \sim \mathcal{N}(\vec{\mu}, R)$ satisfies a n -dimensional **Gaussian distribution**, then the Shannon entropy has explicit form

$$\mathcal{S}(\rho) = \frac{n}{2}(1 + \ln 2\pi) + \frac{1}{2} \ln \det(R).$$

Uncertainty is reflected in the covariance.

Derivation of 1-D case. Let

$$\rho(x) = \frac{1}{\sqrt{2\pi R}} \exp\left(-\frac{(x - \mu)^2}{2R}\right).$$

Then the Shannon entropy becomes

$$\begin{aligned} - \int \rho(x) \ln \rho(x) dx &= - \int \rho(x) \left(-\frac{1}{2} \ln(2\pi R) - \frac{(x - \mu)^2}{2R} \right) dx \\ &= \frac{1}{2} \ln(2\pi R) + \frac{1}{2R} \int (x - \mu)^2 \rho(x) dx \\ &= \frac{1}{2} \ln(2\pi) + \frac{1}{2} \ln R + \frac{1}{2}. \end{aligned}$$

- Specifically, we take $\rho := \rho(U_t) = \rho(u_t - u_t^M)$, where u_t and u_t^M are the perfect and imperfect model output, respectively, with the same mean value. Then,

$$\mathcal{S}(\rho) = \frac{n}{2}(1 + \ln 2\pi) + \frac{1}{2} \ln \det \left(U_t \otimes U_t^* \right).$$

- The definition of the *root-mean-square (RMS) error* is given by

$$\left(\text{RMSE}(U_t) \right)^2 = \frac{1}{T} \int_0^T U_t \otimes U_t^* dt$$

- Take $t \rightarrow \infty$ and assume the expectation value of U_∞ is zero. Then

$$\mathcal{S}(\rho(U_\infty)) = \frac{n}{2}(1 + \ln 2\pi) + \frac{1}{2} \ln \det \left(\text{RMSE}(U_\infty) \right)^2.$$

Thus, the Shannon entropy is the surrogate for the path-wise RMS error.

8. Summary of the Shannon entropy.

- ▶ The Shannon entropy $\mathcal{S}(p)$ of the probability p is defined as

$$\text{Discrete:} \quad \mathcal{S}(p) = \mathcal{S}(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \ln p_i,$$

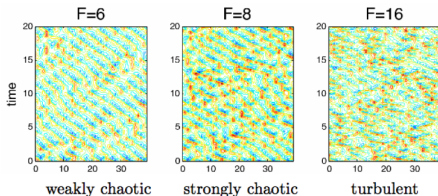
$$\text{Continuous:} \quad \mathcal{S}(\rho) = - \int \rho \ln(\rho) da.$$

which measures the lack of information or the amount of uncertainty.

- ▶ The Shannon entropy is essentially **unique**.
- ▶ **Maximum entropy principle**. The least biased probability distribution p_L given the constraints $\mathcal{C}_L$ is the **maximum entropy distribution**,

$$\max_{p \in \mathcal{C}_L} \mathcal{S}(p) = \mathcal{S}(p_L), \quad p_L \in \mathcal{C}_L.$$

- ▶ **Coarse graining** loses information and increases the entropy.
- ▶ **The Shannon entropy in Gaussian framework is the surrogate for the path-wise RMS error.**



[Output from L-96 model.]

Simple question: How does the system's entropy change from weakly chaotic regime to turbulent one?

Relative entropy

Recall ...

- ▶ Shannon entropy measures the lack of information in one probability distribution function.
- ▶ How to measure the lack of information **in one distribution compared with another?**

Examples:

- ▶ The lack of information (model error) in the imperfect model q compared with nature p .
- ▶ The information gain with additional observations $p(u|y)$ compared with the information only from running the dynamics (climatology) $p(u)$.

1. Motivation – measuring the lack of information in the imperfect model q compared with nature p

- Assume the truth is $p(a)$. The ignorance (lack of information) about a is

$$\mathcal{I}(p(a)) = -\ln(p(a)).$$

The average ignorance, i.e., **the Shannon entropy**, is

$$\langle \mathcal{I}(p(a)) \rangle = - \int p(a) \ln(p(a)) da.$$

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- ▶ Assume the imperfect model is $p^M(a)$. The ignorance in the model about a is

$$\mathcal{I}(p^M(a)) = -\ln(p^M(a)).$$

The model's expected ignorance is

$$\langle \mathcal{I}(p^M(a)) \rangle = - \int p(a) \ln(p^M(a)) da.$$

Note here **the outcomes are actually generated by p .**

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$$\langle \mathcal{I}(p^M(a)) \rangle = - \int p(a) \ln(p^M(a)) da.$$

Note here the outcomes are actually generated by p .

- ▶ Consider now,

$$\begin{aligned} \langle \mathcal{I}(p^M(a)) \rangle - \langle \mathcal{I}(p(a)) \rangle &= \int p(a) \log(p(a)) da - \int p(a) \ln(p^M(a)) da \\ &= \int p(a) \ln \frac{p(a)}{p^M(a)} da = \mathcal{P}(p, p^M). \end{aligned}$$

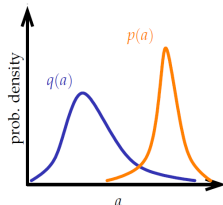
Interpretation: $\mathcal{P}(p, p^M)$ is an objective metric for model error that measures the expected increase in ignorance, or lack of information, about the system incurred by using p^M , when the outcomes are actually generated by p .

2. Definition².

The relative entropy of q compared with p is given by

$$\mathcal{P}(p, q) = \int p \ln \frac{p}{q}.$$

- ▶ $\mathcal{P}(p, q)$ is positive unless $p = q$.
- ▶ $\mathcal{P}(p, q)$ is invariant under general nonlinear change of variables.



However, $\mathcal{P}(p, q)$ is not symmetric under $p \leftrightarrow q$ and does not obey the triangle inequality.

Relative entropy quantifies

- ▶ The model error in the imperfect model q with respect to the perfect one p .
- ▶ The gain of information in the posterior distribution p compared with the prior q .

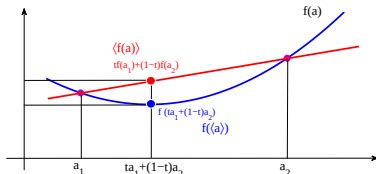
²Also known as Kullback-Leibler (KL) divergence or information divergence.

Proof: Non-negativity of relative entropy

- Jensen's inequality. If f is a convex function and a is a random variable labeled by reals then

$$\langle f(a) \rangle \geq f(\langle a \rangle).$$

Jensen's inequality generalizes the statement that the secant line of a convex function lies above the graph of the function,



$$“\langle f(a) \rangle” = t f(a_1) + (1 - t) f(a_2) \geq f(t a_1 + (1 - t) a_2) = “f(\langle a \rangle)”$$

- Jensen's inequality leads to the fundamental result that $\mathcal{P}(p, q) \geq 0$:

$$\begin{aligned} \mathcal{P}(p, q) &= \int p(a) \ln \frac{p(a)}{q(a)} da \\ &= \left\langle -\ln \frac{q}{p} \right\rangle \\ &\geq -\ln \langle q/p \rangle \quad (\text{Jensen; } -\ln \text{ is convex}) \\ &= -\ln \int p(a) \frac{q(a)}{p(a)} da = 0. \end{aligned}$$

Proof: Invariance under general nonlinear change of variable

Assume a change of variable from x to y . Then according to $p(x)dx = p(y)dy$,

$$\begin{aligned}\mathcal{P}(p(x), q(x)) &= \int p(x) \ln \frac{p(x)}{q(x)} dx = \int p(y) \ln \frac{p(y)dy/dx}{q(y)dy/dx} dy \\ &= \int p(y) \ln \frac{p(y)}{q(y)} dy = \mathcal{P}(p(y), q(y)).\end{aligned}$$

3. maximum relative entropy principle

The least biased probability measure p^* , given all the constraints $\mathcal{C}$ and external bias, is the one that satisfies

$$\max_{p \in \mathcal{C}} \mathcal{S}(p, p_0) = \mathcal{S}(p^*, p_0),$$

where

$$\mathcal{S}(p, p_0) = - \sum_{i=1}^n p_i \ln \left(\frac{p_i}{p_i^0} \right).$$

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Proposition 1

If there is no real external bias, then the maximum entropy principle should yield the same probability distribution p^* as the maximum entropy principle.

In fact, without external bias,

$$p_0 = \sum_{i=1}^n \frac{1}{n} \delta_{a_i}.$$

Then the relative entropy $\mathcal{S}(p, p_0)$ differs from the entropy $\mathcal{S}(p)$ by a constant,

$$\mathcal{S}(p, p_0) = \mathcal{S}(p) - \ln n.$$

3. maximum relative entropy principle

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where

$$\mathcal{S}(p, p_0) = - \sum_{i=1}^n p_i \ln \left(\frac{p_i}{p_i^0} \right).$$

Proposition 2

If the only constraint on the system is given by the external bias p_0 , then the probability distribution predicted by the maximum relative entropy principle must be the external bias itself, $p^* = p_0$.

In this case the only restricted condition is $\sum_{i=1}^n p_i = 1$. Utilizing the Lagrange multiplier,

$$-\nabla_p \mathcal{S}|_{p=p^*} + \lambda \nabla_p \left(\sum_{i=1}^n p_i \right) |_{p=p^*} = 0,$$

and componentwise it yields

$$\ln \left(\frac{p_i^*}{p_i^0} \right) + (1 + \lambda) = 0, \quad i = 1, \dots, n.$$

This in turn implies that the ratio p_i/p_i^0 is a constant independent of i and thus $p^* = p_0$.

4. Empirical theory for model selection

- ▶ Recall the maximum entropy distribution associated with L measurement

$$p_L(a) = \exp \left(-\alpha_0 - \sum_{j=1}^L \alpha_j E_j(a) \right).$$

- ▶ In practice, p^M is determined by no more information than that available in the observations. Thus, it is reasonable to assume that

$$p^M(a) := p_{L'}^M(a) = \exp \left(-\mu_0 - \sum_{j=1}^{L'} \mu_j E_j(a) \right), \quad L' \leq L.$$

- ▶ Key fact:

$$\mathcal{P}(p, p^M) = S(p_L) - S(p) + \mathcal{P}(p_L, p^M),$$

or

$$\mathcal{P}(p, p^M) = \mathcal{P}(p, p_L) + \mathcal{P}(p_L, p^M).$$

- ▶ $S(p_L) - S(p)$ measures the **intrinsic error** from incomplete measurement of nature's signal – unknown in practice. It is the **information barrier**.
- ▶ $\mathcal{P}(p_L, p^M)$ can be calculated by using **observed** p_L from true signal.
- ▶ **Model selection strategy:** Among a class of imperfect models, $\mathcal{M} = \{M_1, M_2, \dots\}$, select the minimum lack of information model, given by

$$\min_{M \in \mathcal{M}} \mathcal{P}(p_L, p^M) = \min_{M \in \mathcal{M}} \mathcal{P}(p, p^M).$$

Now we prove

$$\mathcal{P}(p, p^M) = \mathcal{S}(p_L) - \mathcal{S}(p) + \mathcal{P}(p_L, p^M).$$

► We first establish that

$$\int p_L(a) \ln p^M(a) da = \int p(a) \ln p^M(a) da.$$

Proof:

$$\begin{aligned} \int p_L(a) \ln p^M(a) da &= \int p_L(a) \left(-\mu_0 - \sum_{j=1}^{L'} \mu_j E_j(a) \right) da \\ &= -\mu_0 - \sum_{j=1}^{L'} \mu_j \langle E_j(a) \rangle \\ &= -\mu_0 \int p(a) da - \sum_{j=1}^{L'} \mu_j \int p(a) E_j(a) da \\ &= \int p(a) \left(-\mu_0 - \sum_{j=1}^{L'} \mu_j E_j(a) \right) da \\ &= \int p(a) \ln p^M(a) da. \end{aligned}$$

- With the previous result, we have

$$\begin{aligned}\mathcal{P}(p, p^M) &= \int p(a) \ln \frac{p(a)}{p^M(a)} da \\&= \int p(a) \ln p(a) da - \int p(a) \ln p^M(a) da \\&= -S(p) - \int p_L(a) \ln p^M(a) da \quad \text{(using previous result)} \\&= -S(p) + \int p_L(a) \ln \frac{p_L(a)}{p^M(a)} da - \int p_L(a) \ln p_L(a) da \\&= S(p_L) - S(p) + \mathcal{P}(p_L, p^M).\end{aligned}$$

- With the previous result, we have

$$\begin{aligned}
 \mathcal{P}(p, p^M) &= \int p(a) \ln \frac{p(a)}{p^M(a)} da \\
 &= \int p(a) \ln p(a) da - \int p(a) \ln p^M(a) da \\
 &= -S(p) - \int p_L(a) \ln p^M(a) da \quad (\text{using previous result}) \\
 &= -S(p) + \int p_L(a) \ln \frac{p_L(a)}{p^M(a)} da - \int p_L(a) \ln p_L(a) da \\
 &= S(p_L) - S(p) + \mathcal{P}(p_L, p^M).
 \end{aligned}$$

- Finally, making use of $p_L(a) = \exp \left(-\alpha_0 - \sum_{j=1}^L \alpha_j E_j(a) \right)$, we have

$$\begin{aligned}
 \mathcal{P}(p, p_L) &= - \int p(a) \ln p(a) da - \int p(a) \ln p_L(a) da \\
 &= -S(p) + \alpha_0 \int p(a) da + \sum_{j=1}^L \alpha_j \int E_j(a) p(a) da \\
 &= -S(p) + \alpha_0 \int p_L(a) da + \sum_{j=1}^L \alpha_j \int E_j(a) p_L(a) da \\
 &= -S(p) + S(p_L).
 \end{aligned}$$

5. Additional observational constraints

Suppose that additional observations come in, so that

$$\mathcal{S}(p_L) < \mathcal{S}(p_{L'}), \quad \text{with } L > L'.$$

From our fundamental relation,

$$\begin{aligned} \mathcal{P}(p, p_{L'}^M) &= \mathcal{S}(p_L) - \mathcal{S}(p) + \mathcal{P}(p_L, p_{L'}^M) \\ &= \mathcal{S}(p_{L'}) - \mathcal{S}(p) + \mathcal{P}(p_{L'}, p_{L'}^M), \end{aligned}$$

it follows that

$$\mathcal{P}(p_L, p_{L'}^M) > \mathcal{P}(p_{L'}, p_{L'}^M).$$

This implies

- ▶ Lack of information (uncertainty) increases in the imperfect models with additional observational constraints.
- ▶ Models must be recalibrated when new observational constraints become available.

6. Estimating the error in posterior distributions

- ▶ In many applications of interest, we are encountered with the following situation:
 - ▶ u : state variable (inaccessible).
 - ▶ y : incomplete, noisy observations.
 - ▶ $p(u|y)$: conditional distribution of u given that we have observed y .
- ▶ Our **goal** is to compare $p(u|y)$ with the posterior distribution $p_{\theta}^M(u|y)$ produced by an imperfect model that depends on parameters θ .

Bayesian setup

$$\begin{aligned} p(u|y) p(y) &= p(u, y) = p(y|u) \pi(u), \\ p_{\theta}^M(u|y) p_{\theta}^M(y) &= p_{\theta}^M(u, y) = p_{\theta}^M(y|u) \pi_{\theta}^M(u). \end{aligned}$$

- ▶ $\pi(u), \pi_{\theta}^M(u)$: prior distributions.
- ▶ $p(y|u), p_{\theta}^M(y|u)$: observation processes, e.g., $p(y|u) \sim \mathcal{N}(\mu(u), \sigma^2)$ for Gaussian noise.
- ▶ $p(u|y), p_{\theta}^M(u|y)$: posterior distributions.
- ▶ Densities of states:

$$\begin{aligned} p(y) &= \int p(u, y) du = \int p(y|u) \pi(u) du, \\ p_{\theta}^M(y) &= \int p_{\theta}^M(u, y) du = \int p_{\theta}^M(y|u) \pi_{\theta}^M(u) du. \end{aligned}$$

- The *conditional relative entropy*,

$$\bar{\mathcal{P}}\left(p(u|y), p_{\theta}^M(u|y)\right) = \int p(y) \mathcal{P}\left(p(u|y), p_{\theta}^M(u|y)\right) dy,$$

measures the expected lack of information (w.r.t. $p(y)$) in the posterior distribution of the imperfect model over all observations y . It is referred to as a **superensemble average**.

- The following is an identity:

$$\mathcal{P}(p(u, y), p_{\theta}^M(u, y)) = \mathcal{P}(p(y), p_{\theta}^M(y)) + \bar{\mathcal{P}}(p(u|y), p_{\theta}^M(u|y)).$$

- If the observation model is consistent with nature,

$$p_{\theta}^M(y|u) = p(y|u),$$

then

$$\mathcal{P}(p(u, y), p_{\theta}^M(u, y)) = \mathcal{P}(\pi(u), \pi_{\theta}^M(u)),$$

leading to the bound

$$\begin{aligned} \bar{\mathcal{P}}(p(u|y), p_{\theta}^M(u|y)) &= \mathcal{P}(\pi(u), \pi_{\theta}^M(u)) - \mathcal{P}(p(y), p_{\theta}^M(y)) \\ &\leq \mathcal{P}(\pi(u), \pi_{\theta}^M(u)). \end{aligned}$$

- Lack of information in the posterior is bounded by that in the prior.

7. Explicit calculation of the observable model error

Assume the perfect model and imperfect model has the following expressions:

$$p_L(a) = e^{-\alpha_0} e^{-\sum_{j=1}^L \alpha_j E_j(a)}, \quad p^M(a) = e^{-\mu_0} e^{-\sum_{j=1}^{L'} \mu_j E_j(a)},$$

we have

$$\begin{aligned} \mathcal{P}(p_L, p^M) &= \int p_L(a) \left(\ln p_L(a) da - \ln p^M(a) \right) da \\ &= \int p_L(a) \left(\left(-\alpha_0 - \sum_{j=1}^L \alpha_j E_j(a) \right) - \left(-\mu_0 - \sum_{j=1}^{L'} \mu_j E_j(a) \right) \right) da \\ &= -(\alpha_0 - \mu_0) - \sum_{j=1}^L (\alpha_j - \mu_j) \bar{E}_j, \end{aligned}$$

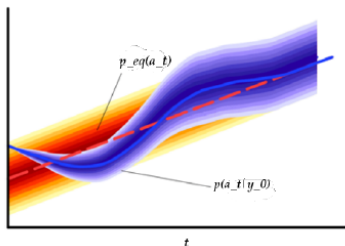
where $\bar{E}_j$ are the measurements/observations:

$$\bar{E}_j = \int p_L(a) E_j(a) da = \int p(a) E_j(a) da.$$

If we know $\{\alpha_j\}$ and $\{\mu_j\}$, it is possible to compute the observable error without knowing the true distribution.

8. Information-theoretic measures of predictive skill and model error.

- ▶ Consider that the statistics of the true signal $u(t)$ at the forecast lead time t given initial data $u_0 = u(0)$ are described by a time-dependent PDF $p_t(u|u_0)$ with a well-defined equilibrium $p_{eq}(u) = \lim_{t \rightarrow \infty} p_t(u|u_0)$.
- ▶ The imperfect model produces forecast PDFs $p_t^M(u|u_0)$ and the equilibrium is given by $p_{eq}^M(u) = \lim_{t \rightarrow \infty} p_t^M(u|u_0)$, but these PDFs will in general differ from the PDFs of the true signal.



- **Intrinsic predictability, $\mathcal{D}_t$.** This metric quantifies the information provided by the initial conditions about the future state of the system beyond the prior knowledge available through equilibrium statistics,

$$\mathcal{D}_t = \mathcal{P}(p_t, p_{eq}).$$

- **Internal predictive skill, $\mathcal{D}_t^M$.** In direct analogy with the intrinsic predictability, the internal predictive skill measures information gain in the time-dependent forecast PDF of the imperfect model beyond the model's equilibrium,

$$\mathcal{D}_t^M = \mathcal{P}(p_t^M, p_{eq}^M).$$

Note that $\mathcal{D}_t^M$ will convey false predictability if either of p_t^M or p_{eq}^M are biased away from the truth.

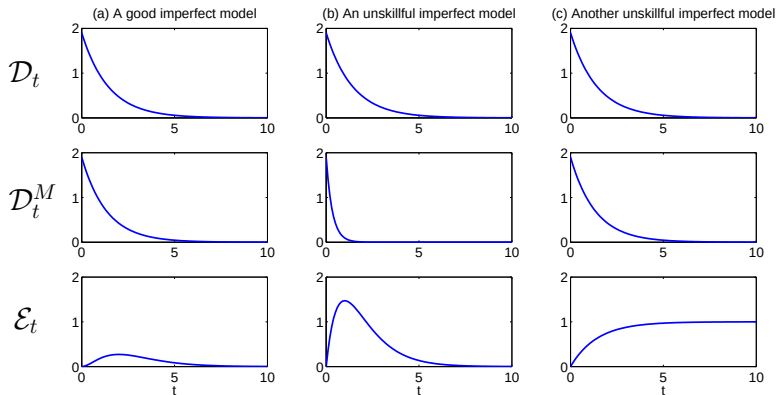
Nevertheless, it is an important metric for model assessment since it measures potential initial value predictive skill in the model relative to a trivial forecast drawn from the model's equilibrium.

- **Model error, $\mathcal{E}_t$.** This metric measures the lack of information in the imperfect model density compared to a perfect statistical forecast,

$$\mathcal{E}_t = \mathcal{P}(p_t, p_t^M)$$

Examples.

- ▶ (a) good case
- ▶ (b) large error in model sensitivity
- ▶ (c) large error in model fidelity



9. Special case: Gaussian distribution

Recall that the relative entropy of q compared with p is given by

$$\mathcal{P}(p, q) = \int p \ln \frac{p}{q}.$$

When both $p \sim \mathcal{N}(\vec{m}_p, R_p)$ and $q \sim \mathcal{N}(\vec{m}_q, R_q)$ are Gaussian, the relative entropy has an explicit formula,

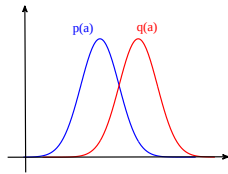
$$\mathcal{P}(p, q) = \underbrace{\left[\frac{1}{2} (\vec{m}_p - \vec{m}_q)^T R_q^{-1} (\vec{m}_p - \vec{m}_q) \right]}_{\text{Signal}} + \underbrace{\frac{1}{2} \left[\text{tr}(R_p R_q^{-1}) - |\mathbf{K}| - \ln \det(R_p R_q^{-1}) \right]}_{\text{Dispersion}},$$

where $|\mathbf{K}|$ is the dimension of both the distribution.

- ▶ Signal measures the lack of information in the **mean** weighted by model covariance.
- ▶ Dispersion involves the **covariance** ratio.

This also implies that even in the Gaussian framework, the uncertainty reduction **is not merely reflected in the variance!**

[Compare with the Shannon entropy.]



10. Summary of the relative entropy

- ▶ The relative entropy $\mathcal{P}(p, q)$ measures the **lack of information** about the system incurred by using q , when the outcomes are actually generated by p .
 - ▶ $\mathcal{P}(p, q) \geq 0$.
 - ▶ $\mathcal{P}(p, q)$ is invariant under general nonlinear change of variables.
- ▶ Empirical theory for model selection:

$$\mathcal{P}(p, p^M) = \underbrace{\mathcal{P}(p, p_L)}_{\text{intrinsic barrier}} + \underbrace{\mathcal{P}(p_L, p^M)}_{\text{model error}}.$$

Model selection criteria: $\min_{M \in \mathcal{M}} \mathcal{P}(p_L, p^M) = \min_{M \in \mathcal{M}} \mathcal{P}(p, p^M).$

- ▶ Lack of information (uncertainty) increases in the imperfect models with additional observational constraints

$$\mathcal{P}(p_L, p_{L'}^M) > \mathcal{P}(p_{L'}, p_{L'}^M) \quad L > L'.$$

- ▶ Lack of information in the posterior is bounded by that in the prior,

$$\begin{aligned} \bar{\mathcal{P}}(p(u|y), p_\theta^M(u|y)) &= \mathcal{P}(\pi(u), \pi_\theta^M(u)) - \mathcal{P}(p(y), p_\theta^M(y)) \\ &\leq \mathcal{P}(\pi(u), \pi_\theta^M(u)). \end{aligned}$$

- ▶ In Gaussian framework

$$\mathcal{P}(p, q) = \underbrace{\left[\frac{1}{2} (\vec{m}_p - \vec{m}_q)^T R_q^{-1} (\vec{m}_p - \vec{m}_q) \right]}_{\text{Signal}} + \underbrace{\frac{1}{2} \left[\text{tr}(R_p R_q^{-1}) - |\mathbf{K}| - \ln \det(R_p R_q^{-1}) \right]}_{\text{Dispersion}}.$$

Mutual information

1. Definition

- ▶ The mutual information³ of two random variables is a measure of the variables' mutual dependence,

$$\mathcal{M}(X, Y) = \int_X \int_Y p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} dx dy.$$

- ▶ Relation with the relative entropy:

$$\mathcal{M}(X, Y) = \mathcal{P}(p(x, y), p(x)p(y)),$$

and thus $\mathcal{M}(X, Y) > 0$ unless X and Y are independent random variables.

³also known as transinformation

- Relation with the Shannon entropy:

$$\begin{aligned}
 \mathcal{M}(X, Y) &= \int_X \int_Y p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} dx dy \\
 &= \int_X \int_Y p(x, y) \ln \frac{p(x|y)}{p(x)} dx dy \\
 &= - \int_X \int_Y p(x, y) \log p(x) dx dy + \int_X \int_Y p(x, y) \ln p(x|y) dx dy \\
 &= - \int_X p(x) \ln p(x) dx - \left(- \int_X \int_Y p(x, y) \ln p(x|y) dx dy \right) \\
 &= S(p(x)) - S(p(x|y)) := S(X) - S(X|Y),
 \end{aligned}$$

The conditional entropy: $S(X|Y) = \int_X \int_Y p(x, y) \ln p(x|y) dx dy$

The joint entropy: $S(X, Y) = \int_X \int_Y p(x, y) \ln p(x, y) dx dy.$

- We have the following,

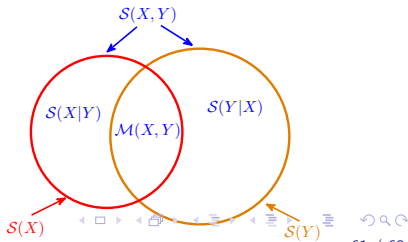
$$\mathcal{M}(X, Y) = \mathcal{M}(Y, X),$$

$$\mathcal{M}(X, Y) = S(X) - S(X|Y),$$

$$\mathcal{M}(X, Y) = S(Y) - S(Y|X),$$

$$\mathcal{M}(X, Y) = S(X) + S(Y) - S(X, Y),$$

$$\mathcal{M}(X, X) = S(X).$$



2. Superensemble revisited

- ▶ Recall that $p(u)$ is the prior distribution and $p(u|y)$ is the posterior distribution with observations y .
- ▶ The additional information beyond $p(u)$ gained by having observed y is given by

$$D^y = \mathcal{P}\left(p(u|y), p(u)\right) = \int_U p(u|y) \ln \frac{p(u|y)}{p(u)} du.$$

- ▶ The superensemble expectation value of D_t^y is given by

$$\begin{aligned} D &= \int_Y D^y p(y) dy = \int_Y \left(\int_U p(u|y) \ln \frac{p(u|y)}{p(u)} du \right) p(y) dy \\ &= \int_U \int_Y p(u|y) p(y) \ln \frac{p(u|y)p(y)}{p(u)p(y)} du dy \\ &= \int_U \int_Y p(u, y) \ln \frac{p(u, y)}{p(u)p(y)} du dy \\ &= \mathcal{S}\left(p(u, y), p(u)p(y)\right) = \mathcal{M}(U, Y). \end{aligned}$$

This implies

- ▶ If u and y are almost independent, then $\mathcal{M}(U, Y)$ is small; little information is gained with observations.
- ▶ If u strongly depends on y , then $\mathcal{M}(U, Y)$ is large; a significant amount of information is gained in posterior with observations.

3. Gaussian framework

$$\mathcal{M}(X, Y) = \int_X \int_Y p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} dx dy = \mathcal{P}(p(x, y), p(x)p(y)).$$

Assume $x \sim \mathcal{N}(\vec{m}_x, R_x)$ and $y \sim \mathcal{N}(\vec{m}_y, R_y)$ are both n -dimensional.

- Recall the relative entropy in Gaussian framework

$$\begin{aligned} \mathcal{P}(\tilde{p}, \tilde{q}) &= \left[\frac{1}{2} (\vec{m}_{\tilde{p}} - \vec{m}_{\tilde{q}})^T R_{\tilde{q}}^{-1} (\vec{m}_{\tilde{p}} - \vec{m}_{\tilde{q}}) \right] && \text{signal} \\ &+ \frac{1}{2} \left[\text{tr}(R_{\tilde{p}} R_{\tilde{q}}^{-1}) - 2n - \ln \det(R_{\tilde{p}} R_{\tilde{q}}^{-1}) \right] && \text{dispersion.} \end{aligned}$$

Here $\tilde{p} = p(x, y)$ and $\tilde{q} = p(x)p(y)$.

- The mean values of $p(x, y)$ and $p(x)p(y)$ are the same and thus the signal part disappears. The first and second terms in the dispersion part cancel due to

$$\text{tr}(R_{\tilde{p}} R_{\tilde{q}}^{-1}) = \text{tr} \left(\begin{pmatrix} R_x & R_{xy} \\ R_{yx} & R_y \end{pmatrix} \begin{pmatrix} R_x^{-1} & 0 \\ 0 & R_y^{-1} \end{pmatrix} \right) = 2n.$$

- Thus,

$$\mathcal{M}(X, Y) = -\frac{1}{2} \ln \det(R_{\tilde{p}} R_{\tilde{q}}^{-1}) = \frac{1}{2} \ln \left(\frac{\det(R_x) \det(R_y)}{\det(R)} \right),$$

where

$$R = \begin{pmatrix} R_x & R_{xy} \\ R_{yx} & R_y \end{pmatrix}.$$

In Gaussian framework,

$$\mathcal{M}(X, Y) = -\frac{1}{2} \ln \det(R_{\tilde{p}} R_{\tilde{q}}^{-1}) = \frac{1}{2} \ln \left(\frac{\det(R_x) \det(R_y)}{\det(R)} \right),$$

where

$$R = \begin{pmatrix} R_x & R_{xy} \\ R_{yx} & R_y \end{pmatrix}.$$

- Consider 1-dimensional case ($n = 1$),

$$\begin{aligned} \mathcal{M}(X, Y) &= \frac{1}{2} \ln \left(\frac{R_x R_y}{R_x R_y - R_{xy}^2} \right) \\ &= \ln \left(\sqrt{\frac{1}{1 - r^2}} \right), \end{aligned}$$

where the correlation coefficient $r = R_{xy} / (R_x R_y)$.

The mutual information increases when the correlation between x and y increases. **Thus, the mutual information is the surrogates for the path-wise pattern correlation.**

4. Summary

- ▶ The mutual information of two random variables is a measure of the variables' mutual dependence,

$$\mathcal{M}(X, Y) = \int_X \int_Y p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} dx dy.$$

It has the relation with the relative entropy:

$$\mathcal{M}(X, Y) = \mathcal{P}(p(x, y), p(x)p(y)),$$

- ▶ The **superensemble expectation value** of the information gain with observations D^y is the mutual information between state variable and observations D ,

$$D^y = \mathcal{P}(p(u|y), p(u)) = \int_U p(u|y) \ln \frac{p(u|y)}{p(u)} du,$$

$$D = \int_Y D^y p(y) dy = \mathcal{M}(U, Y).$$

- ▶ We prove that **the mutual information in Gaussian framework is the surrogates for the path-wise pattern correlation.**

Let's answer the questions proposed at the beginning ...

- ▶ What is the intrinsic uncertainty in a system described by a probability distribution p ?

Answer: The Shannon entropy $S(p)$.

- ▶ What is the model error in an imperfect model p^M compared with nature p ?

Answer: The relative entropy $\mathcal{P}(p, p^M)$.

- ▶ What is the expected amount of information gained about the system state u via an incomplete measurement y ?

Answer: The mutual information $\mathcal{M}(u; y)$.

Questions and comments?

- ▶ Basic concepts of information theory and the Shannon entropy

1) The Shannon entropy, 2) uniqueness, 3) maximum entropy principle, 4) coarse graining and loss of information, 5) Gaussian framework and the relation with the RMS error

- ▶ The relative entropy

1) The relative entropy, 2) quantifying the lack of information, 3) empirical theory for model selection, 4) estimating errors in posterior distribution, 5) Gaussian framework

- ▶ The mutual information

1) The mutual information, 2) superensemble expectation, 3) Gaussian framework and the relation with the pattern correlation

Thank you!

References and acknowledgements

1. A. J. Majda and X. Wang, Nonlinear Dynamics and Statistical Theories for Basic Geophysical Flows, Cambridge University Press, 2006 (especially Chapter 6).
2. Majda, Andrew, and Nan Chen, Model error, information barriers, state estimation and prediction in complex multiscale systems. Entropy (2018)
3. We gratefully acknowledge that part of these lecture notes is adapted from Nan Chen's original materials and other online resources.